

$$(z-a) \rightarrow 0 \quad (z-a)f(z) \rightarrow k$$

$$\oint_{\gamma} f(z) dz = i k (\theta_2 - \theta_1)$$

$$z \rightarrow \infty \quad z f(z) \rightarrow k \quad \int_{C_R} f(z) dz = -i k (\theta_2 - \theta_1)$$

$$I = \int_0^{2\pi} R(\sin \theta, \cos \theta) d\theta \quad \cos \theta = \frac{z+1/z}{2} \quad \sin \theta = \frac{z-1/z}{2i} \quad d\theta = \frac{dz}{iz}$$

$$J = \int_{-\infty}^{+\infty} f(x) dx \quad z f(z) \rightarrow 0$$

$$I = \int_{-\infty}^{+\infty} f(x) \cos px dx \quad \sin px$$

Jordan $Q(z) \rightarrow 0 \quad \int_{C_R} Q(z) e^{i p z} dz = 0$

$$\int_0^{2\pi} \cos^{2n} \theta d\theta$$

$$\cos \theta = \frac{z+1/z}{2} \quad d\theta = \frac{dz}{iz}$$

$$= \frac{1}{i} \int_{|z|=1} \left(\frac{z^2+1}{2z} \right)^{2n} \cdot \frac{1}{z} dz$$

$$= \frac{1}{i} \frac{1}{2^{2n}} \int_{|z|=1} \frac{(z^2+1)^{2n}}{z^{2n+1}} dz \rightarrow z^{2n}$$

$z=0$, pole, order is $2n+1$

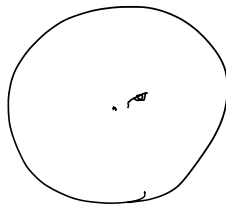
$$\frac{a-1}{(z^2+1)^{2n}} = \sum_{k=0}^{\infty} \binom{k}{2n}$$

$$\downarrow \quad \frac{z^{4n} + z^{4n-2}}{\sum_{k=0}^{\infty} \binom{k}{2n} x^k} \quad x = z^2$$

$$\frac{(2n)!}{n!n!} \quad \frac{(2n)!}{k!(2n-k)!}$$

$$\frac{\pi}{2^{2n-1}} \frac{(2n)!}{n!n!}$$

$$(2) \quad \int_0^{2\pi} \frac{dx}{(a+b \cos x)^2}$$



$$(3) \quad \int_0^{\pi} \frac{d\theta}{1+\sin^2 \theta} \quad \frac{1}{2} \int_0^{2\pi} \frac{d\theta}{1+\sin^2 \theta}$$

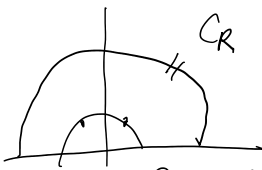
$$\downarrow \quad \cos 2\theta \quad \frac{1}{\phi} \rightarrow \int_0^{2\pi} \frac{d\phi}{3-\cos \phi}$$

$$(4) \quad \int_0^{\pi} \frac{d\theta}{(1+\sin^2 \theta)^2} = \int_0^{2\pi} \frac{z d\phi}{(3-\cos \phi)^2}$$

6: $1+\cos x^2$, C_R

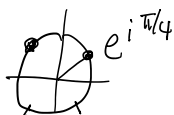
6. $\int_{-\infty}^{+\infty} \frac{x^2}{1+x^4} dx$

$\int_C \frac{z^2}{1+z^4} dz = \int_{-R}^R \frac{x^2}{1+x^4} dx + \int_{C_R} \frac{z^2}{1+z^4} dz \xrightarrow{R \rightarrow \infty} 0$



$= 2\pi i \sum_k \text{res } f(b_k)$

$1+z^4$

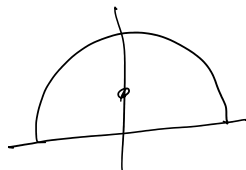


poles $e^{i\pi/4} \quad e^{i3\pi/4}$

z^2

$(z - e^{i\pi/4})(z - e^{i3\pi/4})(z - e^{i5\pi/4})(z - e^{i7\pi/4})$

② $\int_{-\infty}^{+\infty} \frac{1}{(1+x^2)^{n+1}} dx$



$1+z^2 \quad z=i, -i \quad n+1$

$a_1 = \frac{d^{n-1}}{dz^{n-1}} \left[\frac{(z-b)^m f(z)}{(z-i)^{n+1}(z+i)^{n+1}} \right] \Big|_{z=b} \cdot \frac{1}{(m-1)!}$

ordern

$\frac{d^n}{dz^n} \frac{1}{(z+i)^{n+1}} \Big|_{z=i}$

③ $\int_{-\infty}^{+\infty} \frac{x^{2m}}{1+x^{2n}} dx$

$n > m$



$1+z^{2n}$

$z^{2n} = -1 = e^{+i(\pi+2k\pi)}$

$e^{i(k+\frac{1}{2})\pi/n} = b_k$

$(z-b_1)(z-b_2) \dots (z-b_n)$



①

$$\int_{-\infty}^{+\infty} f(x) \cos px \, dx$$

$\sin px$

$$\int_{\mathcal{C}_R} Q(z) e^{ipz} dz \approx 0 \quad Q(z) \rightarrow 0$$

$$\int_0^{\infty} \frac{\cos x}{1+x^4} dx = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos x}{1+x^4} dx$$

$$\frac{1}{2} \int_{\mathcal{C}} \frac{\cos z}{1+z^4} dz = \frac{1}{2} \int_{-\infty}^{+\infty} \frac{\cos x}{1+x^4} dx + \frac{1}{2} \int_{\mathcal{C}_R} \frac{\cos z}{1+z^4} dz$$

$$z^4 = -1 \quad e^{i\pi/4} \quad e^{i3\pi/4}$$

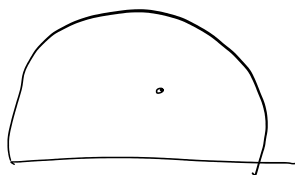
$$\frac{\pi}{2} e^{-1/\sqrt{2}} \cos\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right)$$

$$\int_0^{\infty} \frac{\cos x}{(1+x^2)^3} dx$$

$$z=i \quad z=-i \quad \equiv p_{11}^2$$

$$\frac{1\pi}{16e}$$

$$\textcircled{2} \int_{-\infty}^{+\infty} \frac{x \sin x}{x^2 - 2x + 2} e^{ix}$$

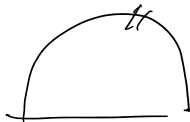


$$(z-1-i)(z-1+i)$$

$$\frac{1+i}{1-i}$$

$$\frac{\sqrt{2}\pi}{e} \sin(1+\pi/4)$$

$$\int_{-\infty}^{+\infty} \frac{x e^{ix}}{x^2 - 2x + 2}$$



$$\textcircled{4} \int_0^{\infty} \frac{\sin(a+2n)x - \sin ax}{(1+x^2) \sin x} \quad \frac{\dots}{(1+x^2)}$$

$$= \frac{\sin(a+2n)x - \sin(a+2n-2)x}{\sin x} \sum_{k=0}^{n-1} \int_0^{\infty} \frac{\cos(a+2k+1)x}{1+x^2} dx =$$

$$\sin(a+2n)x - \sin ax$$

$$\dots (a+2n-2) + \sin(a+2n-2)$$

$$z=i$$

$$\sin(a+2n)x - \sin ax$$

$$= \sin(a+2n)x - \sin(a+2n-2) + \sin(a+2n-2) - \sin(a+2n-4)$$

$$\sin(a+2n-4) - \sin(a+2n-6)$$

$$a_n - a_1$$

$$= a_n - a_{n-1}$$

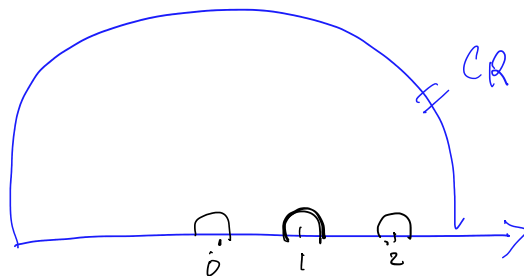
$$+ a_{n-1} - a_{n-2}$$

$$\dots$$

$$+ a_2 - a_1$$

$$\sum_{k=0}^{n-1} \pi e^{-(a+2k+1)} = \pi e^{-a-1} \frac{1-e^{-2n}}{1-e^{-2}}$$

$$\int_{-\infty}^{+\infty} \frac{dx}{x(x-1)(x-2)}$$

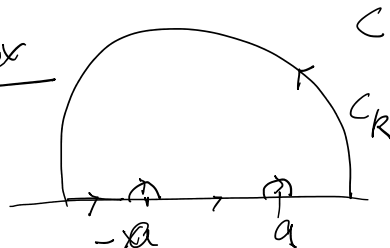


$$(z-a)f(a) \rightarrow K \quad \int_{C_\delta} f(z) dz = \underline{iK(\theta_2 - \theta_1)}$$

$$(z-1) \frac{1}{z(z-1)(z-2)} \rightarrow \frac{1}{1(1-2)} = -1$$

$$\textcircled{2} \quad \int_0^\infty \frac{\sin(x+a)\sin(x-a)}{x^2-a^2} dx$$

$$= \frac{1}{4} \int_0^\infty \frac{\cos 2a - \cos 2x}{x^2-a^2} dx$$



$$0 = \int_C \frac{\cos 2a - \cos 2z}{z^2 - a^2} dz$$

$$= \int_{C_R} \frac{\cos 2a - \cos 2z}{z^2 - a^2} dz \rightarrow 0$$

$$= \int_{-\infty}^{\infty} \frac{\cos 2a - \cos 2x}{x^2 - a^2} dx = \int \frac{\cos 2a - e^{i2z}}{x^2 - a^2} dz$$

$$+ \text{v.p.} \int_{-\infty}^{+\infty} \frac{\cos 2a - \cos 2x}{x^2 - a^2} dx - \int_{C_{\delta, -a}} \frac{\cos 2a - e^{i2z}}{(z-a)(z+a)} dz$$

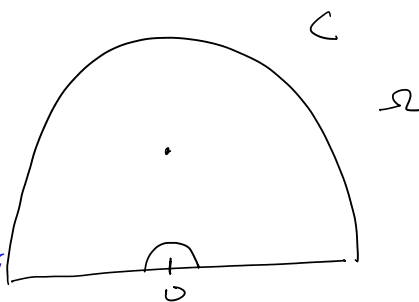
$$- \int_{C_{\delta, a}} \frac{\cos 2a - e^{i2z}}{(z-a)(z+a)} dz$$

$$\lim_{z \rightarrow b} (z-b) f(z) = \lim_{z \rightarrow a} \frac{\cos 2a - e^{i2z}}{z+a} = \frac{\cos 2a - e^{i2a}}{2a} \cdot i \cdot \pi$$

$$\frac{\pi}{4a} \sin 2a$$

$$(3) \int_0^{\infty} \frac{x - \sin x}{x^3(1+x^2)} dx$$

$$= \text{Im} \int_0^{\infty} \frac{x - e^{ix}}{x^3(1+x^2)} dx$$



$$\int_C \frac{z - \sin z}{z^3(1+z^2)} dz = 2\pi i \cdot \text{res} f(b) \quad b=i$$

$$= 2\pi i \frac{i - \sin i}{i^3(2i)}$$

$$= \int_{C_R} \frac{z - \sin z}{z^3(1+z^2)} dz$$

$$\int_{-\infty}^{+\infty} \frac{x}{x^3(1+x^2)} dx - \left(\int_{-\infty}^{+\infty} \frac{-\sin x}{x^3(1+x^2)} dx \right)$$

$$\int_{-\infty}^{+\infty} \frac{x - \sin x}{x^3(1+x^2)} dx = \int_{-\infty}^{+\infty} \frac{i x - (\cos x + i \sin x)}{x^3(1+x^2)} dx$$

$$\frac{i x - e^{ix}}{x^3(1+x^2)} = \frac{i x - (\cos x + i \sin x)}{x^3(1+x^2)}$$

$$\int_{-\infty}^{+\infty} \frac{i x - e^{ix}}{x^3(1+x^2)} dx$$

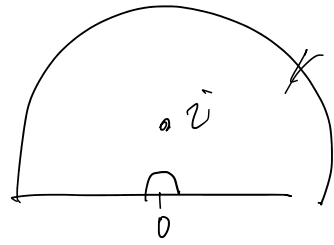
$$z > 0 \text{ (} z > 0 \text{)}$$

$$x > 0 \text{ (} x > 0 \text{)}$$

$$= \text{Im} \left[\int_{-\infty}^{+\infty} \frac{ix - e^{ix} + 1}{x^3(1+x^2)} dx \right]$$

$f(x)$

$f(z)$



$$-i\pi e^{-1} = \int_C \frac{iz - e^{iz} + 1}{z^3(1+z^2)} dz$$

$$= \int_{CR} \frac{iz - e^{iz} + 1}{z^3(1+z^2)} dz + \int_{-\infty}^{+\infty} f(x) dx$$

$$e^x = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots$$

$x = iz$

$$2\pi i \cdot \frac{1 - e^{-1} + 1}{i^3 \cdot (2i)}$$

$$= i\pi(-e^{-1})$$

$$z \cdot f(z) = \frac{(+1) \frac{1}{2} z^2 [1 + z + \dots]}{z^2(1+z^2)}$$

$$z \rightarrow 0$$

$$= +\frac{1}{2}$$

$$\int_{CR} \frac{iz - e^{iz} + 1}{z^3(1+z^2)} dz = \frac{1}{2} \cdot \pi i$$

$$\pi(-e^{-1}) + \frac{1}{2}\pi$$

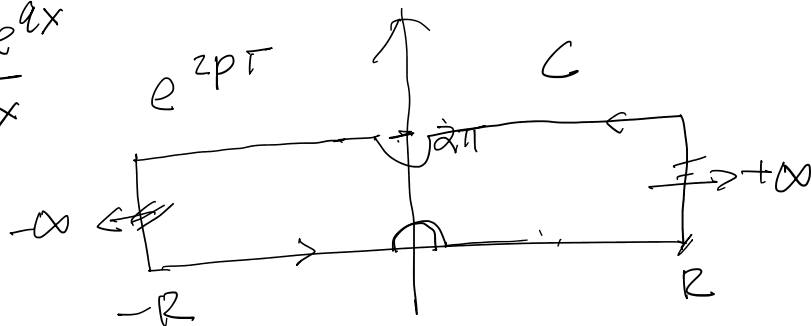
$$(4) \int_{-\infty}^{+\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx$$

analytic

$$0 = \int_C \frac{e^{pz} - e^{qz}}{1 - e^z} dz$$

$$= \int_{-\infty}^{+\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx - \int \frac{e^{pz} - e^{qz}}{1 - e^z} dz$$

$z = R + iy$
 $dz = i dy$



$$= \int_{-\infty}^{+\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx - \int_{(s,0)} \frac{e^z}{1 - e^z} dz \quad dz = i dy$$

$$+ \lim_{R \rightarrow \infty} \int_0^{2\pi} \frac{e^{p(R+iy)} - e^{q(R+iy)}}{1 - e^{R+iy}} i dy \quad \underline{p, q < 1}$$

$$\leq \int_0^{2\pi} \frac{e^{pR} + e^{qR}}{e^R - 1}$$

$$|A| + |B| \geq |A - B| \geq |A| - |B|$$